Michael D. **Catchen**

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Ph.D. candidate with 9+ years of programming experience and 4+ years experience using statistical modeling and machine learning to improve prediction and forecasting in environmental science.

Education

Ph.D. — Dept. of Biology (*expected Feb. 2024*) *Aug. 2020 -*
McGill University *Montréal, QC, CA*
- **Thesis**: *Improving monitoring and prediction of ecological dynamics using simulation and machine learning*
- **Supervisor**: Dr. Andrew Gonzalez
- **Relevant coursework**: Network Science, Probabilistic Programming

M.A. — Dept. Ecology and Evolutionary Biology *2018 - 2020*
University of Colorado Boulder *Boulder, CO, USA*
- **Thesis**: *Phase transitions in landscape connectivity*
- **Supervisor**: Dr. Sam Flaxman
- **Relevant coursework**: Stochastic Processes, Genomics, Evolutionary Ecology, Network Science

B.A. — Dept. of Ecology and Evolutionary Biology *2015 - 2020*
University of Colorado Boulder *Boulder, CO, USA*
- **Relevant coursework**: Algorithms, Data Structures, Applied Probability, Differential Equations, Linear Algebra, Remote Sensing
- **GPA**: 3.92

Publications

A global biodiversity observing system to unite monitoring and guide action *Aug. 24, 2023*
Andrew Gonzalez, Petteri Vihervaara, …, **Michael D. Catchen**, …, Basile van Havre, Elaine Wright **Nature Ecology and Evolution**

Improving ecological connectivity assessments with transfer learning and function approximation *May 5, 2023*
Michael D. Catchen, Michelle Lin, Timothée Poisot, David Rolnick, Andrew Gonzalez **ICLR 2023 — Machine Learning for Remote Sensing**

A roadmap towards predicting species interaction networks (across space and time) *Nov. 8, 2021*
Tanya Strydom*, **Michael D Catchen***, Francis Banville, Dominique Caron, Gabriel Dansereau, Philippe Desjardins-Proulx, Norma R Forero-Muñoz, Gracielle Higino, Benjamin Mercier, Andrew Gonzalez, Dominique Gravel, Laura Pollock, Timothée Poisot **Philosophical Transactions of the Royal Society B**

The missing link: discerning true from false negatives when sampling species interaction networks in review
Michael D Catchen, Timothée Poisot, Laura J Pollock, Andrew Gonzalez in review at **Ecography**

Projecting the rewiring of the Bumblebee-wildflower pollination network of the Southern Rocky Mountains in submission
Michael D Catchen, Paul CaraDonna, Jane E. Ogilvie, Julian Resasco, Timothée Poisot, Laura J. Pollock, Andrew Gonzalez

Professional Experience

Research Software Developer *Dec. 2021 - present*
Group on Earth Obsearvations — Biodiversity Observation Network (GEO-BON) *Montréal, QC, CA*
- Worked on both front and back-end components for a web application for analysis of large-scale geospatial environmental data and planning optimal biodiversity-observation-networks (BONs), as part of a partnership between GEO-BON and Microsoft
- Designed tools for optimizing spatial data sampling design, as well scripts for producing species distribution models using gradient-boosting methods.
- Technologies used: Node.js, React.js, Docker, Julia, PostgreSQL, STAC Catalogues

Flight Software Engineering Intern *May - Aug. 2017*
NASA Jet Propulsion Laboratory *Pasadena, CA, USA*
- Developed flight software for two small satellites (*LunarFlashlight* & *NEAScout*) as part of a medium-sized team
- Wrote code and tests for the interface between the motherboard and inertial-measurement-unit (IMU) of each satellite
- Gained skills in embedded systems programming, unit and integration testing, and software development practices in larger teams

Systems Engineering Intern *May - Aug. 2015, May - Aug. 2016*
NASA Jet Propulsion Laboratory *Pasadena, CA, USA*
- Developed a content management system for documentation of various software components used to uplink commands to spacecraft as part of a medium-sized team

Teaching Experience

Teaching Assistant — General Biology Lab II (EBIO 1240) *Spring 2020*
University of Colorado Boulder *Boulder, CO, USA*

- Taught three sections of 18-20 students in a laboratory environment. Covered basics of statistical analysis in R, evolutionionary biology, phylogenetics, physiology, anatomy, and ecology.

Teaching Assistant — General Biology Lab I (EBIO 1230) *Fall 2019*
University of Colorado Boulder *Boulder, CO, USA*

- Taught three sections of 18-20 students in a laboratory environment. Covered basics of statistical analysis in R, molecular and cellular biology, experimental design, and hypothesis testing

Learning Assistant — Calculus I for Engineers (APPM 1350) *Fall 2017*
Teaching Assistant, University of Colorado Boulder *Boulder, CO, USA*

- Tutored students in small groups, graded exams.

Learning Assistant — Calculus III for Engineers (APPM 2350) *Spring 2017*
Teaching Assistant, University of Colorado Boulder *Boulder, CO, USA*

- Tutored students in small groups, graded exams.

Learning Assistant — Calculus II for Engineers (APPM 1360) *Spring 2016, Fall 2016*
Teaching Assistant, University of Colorado Boulder *Boulder, CO, USA*

- Tutored students in small groups, graded exams.

Invited Talks

False negatives in ecological networks *Aug. 24, 2023*
VERENA Consortium *Washington D.C. (virtual)*

Toward a virtual ecology: Simulating ecosystems to optimize spatial sampling of species interactions *Aug. 15, 2022*
ESA 2022 *Montréal, QC, CA*

Projects

EcoDynamics.jl *2021 - present*
Lead developer

- Developed and maintains a Julia package for simulation of reaction-diffusion population and community dynamics on heterogenous spatial graphs
- Developed intefaces to external Julia packages for Bayesian inference of dynamical systems (using *Turing.jl*) and scientific machine-learning (SciML) with the *DifferentialEquations.jl* and Julia SciML ecosystem.

NeutralLandscapes.jl *2021 - present*
Lead developer

- Maintains Julia package that provides a wide variety of methods for generating landscapes with prescribed statistical properties. Developed novel methods for spatiotemporally autocorrelated change in landscapes, and discrete patch generation.

SpeciesDistributionToolkit.jl *2022 - present*
Contributing developer

- Contributed methods to a Julia package for species distribution modeling, including multivariate transforms of data, performance optimization of pseudoabsence algorithms, and building interfaces to other Julia packages for ecology.

Skills

Quantitative Research	Statistical modeling (spatial/temporal analysis, predictive modeling), Machine learning (Recurrent, Convolutional, and Graph Neural Networks, gradient-boosted tree classifiers), Feature mining and engineering, Model comparison and selection, Numerical simulation, Mathematical optimization
Languages	Julia | Python | R | C++ | JavaScript | MATLAB | SQL | BASH
Tools	PyTorch, PyTorch-Geometric, Flux.jl, Turing.jl, STAN, Git, LaTeX, Github Actions, AWS, scikit-learn, Docker